Fitting Models to Data

Fitting equalibrium model prevelence p^*

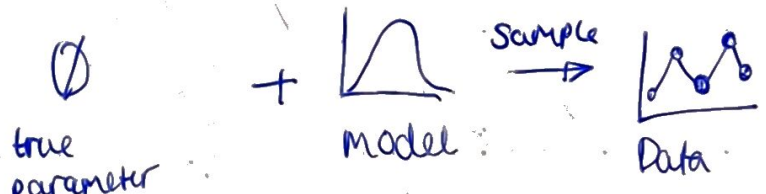
Distance(data, model) = $|x - p|$

Aim: minimise + distance between data & model

Endemic Diseases - Fitting Process:

1. Set up params & model according to prior knowledge
   - Ensure replenishment of susceptibles (demography, waning immunity)
2. Pick β value that results in outbreak
3. Start with normal outbreak conditions
4. Burn-in = Run for long period of time (10 years +)
5. Check tail of outputs - (have values stablized)?
   - If no then keep running
6. Use tail of burn in to set init conditions
7. Use an optimizer to pick values of β that minimises distance

Frequentist Statistics - assumes true parameter value
- Observations are random vars sampled depending on 'true' value



Fitting Results:

- Single parameter estimate
- Some concept of 'goodness of fit'

Conditional probability:

$P[A|B]$ → probability of A when we know B is true

$P[\text{data}|\text{model}]$

Binomial Likelihood:

$P[h \text{ out of } n \mid \#\text{heads} \sim \text{Binomial}(n,p)]$

h → observed heads; n → # throws

$$= \binom{n}{h} p^h \times (1-p)^{n-h}$$

$$= \text{dbinom}(h,n,p)$$ ← R function

Probability of observing data (or result more extreme) $< 1 - 0.95 = 0.05$

↓ p-value

← rejecting the null hypothesis at 95% significance

formula for binominal proportion confidence interval:

$$p \pm z\sqrt{\frac{p(1-p)}{n}}$$

$z = (1-\frac{\alpha}{2})^{th}$ quantile

Bayesian Statistics:

Degree of belief in model / probability

A & B are independant

$p[A|B] = p[A] \times p[B]$ otherwise

$$p[A|B] = \frac{p[A \text{ and } B]}{p[B]}$$

Bayes rule applied to fitting:

$$p[\theta | \text{data}] = \frac{p[\text{data} | \phi]\, p[\phi]}{p[\text{data}]}$$ ← prior beliefs

output of fitting: parameter distribution

likelihood

evidence

Frequentist	Bayesian
+ More objective	- More subjective
+ User friendly	- high dependance on choice of prior
+ less computationally expensive	+ Incorperates prior beliefs
+ estimates based on data	+ outputs are distributed
- p-val & CIs hard to interpret	+ can get measure of how 'good' model is
- All decisions relative to null case	

$\Delta C = \beta S I$ ← cumalative infections